Reviving the Lost Wilderness: Measuring the Educational Value of Immersive Technologies in Prehistoric Wildlife Exploration

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Final Project Report

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

This research addresses the limitations of traditional museum learning environments, which often rely on passive observation and result in low user engagement and limited knowledge retention. To overcome these challenges, an interactive augmented reality–based system, *Fossil Hunt AR*, was developed. The system integrates gamification, state-persistent interaction, and quiz-based validation to transform museum visits into an active learning experience. The application was implemented using Unity and the Vuforia Engine, enabling image-based tracking and real-time interaction within a physical environment. The effectiveness of the system was evaluated through a comparative study involving system-generated telemetry data and a traditional questionnaire-based approach. The results indicate a significant improvement in user engagement, with interaction time increasing by more than 30%, alongside higher task completion rates and evidence of active learning behavior. These findings demonstrate that augmented reality can effectively enhance user engagement and learning outcomes in museum contexts.

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AR	Augmented Reality
VR	Virtual Reality
DSR	Design Science Research
UI	User Interface
UX	User Experience
HCI	Human-Computer Interaction
STEM	Science, Technology, Engineering, and Mathematics
MCQ	Multiple Choice Question
GPS	Global Positioning System
SDK	Software Development Kit
API	Application Programming Interface
3D	Three-Dimensional
2D	Two-Dimensional
LTS	Long-Term Support
GLES	OpenGL for Embedded Systems
JSON	JavaScript Object Notation
FPS	Frames Per Second
ISO	International Organization for Standardization

1. INTRODUCTION

1.1 Background and Literature Survey

The shortcomings of current learning environments in museums have prompted growing interest in the potential of Augmented Reality (AR) technologies to support interactions and enhance engagement. While Virtual Reality (VR) creates an entirely virtual environment in which users are immersed, AR adds digital information to the physical world and users can interact with both physical and digital objects. This makes it an ideal technology for museum settings where there is a need to retain and contextualise physical objects.

In this study, AR is not only used for visualisation but also as a platform for interaction and learning reinforcement. Current museum applications of AR tend to concentrate on the visualisation of 3D models or animations when a marker is detected. Although these features are visually engaging, they do not necessarily support a learning experience. However, the system presented in this study proposes a temporal interaction loop, in which AR is employed to access a range of tasks that engage, explore and confirm learning.

This loop is realised through the use of a treasure-map metaphor, in which visitors are directed to specific locations in the museum. At each location, there is a custom image target (called a "White X") which triggers AR content. This design ensures that interaction is firmly grounded in the physical space of the museum and avoids the possibility of disconnecting interaction with the system from the museum environment.

Technologically, the decision to use image-based tracking with Vuforia is a pragmatic one, given the conditions of deployment. Markerless tracking, although more versatile, is highly dependent on the detection of features in the environment, and demands more computational power. In indoor environments, particularly in museums with variable lighting and low texture contrast, this can lead to poor tracking and user experience. On the other hand, using pre-defined image targets offers high accuracy, stability, and predictability, with AR content accurately placed relative to the physical objects.

The system design also integrates the use of tools for interaction, such as a shovel and brush for excavation and cleaning. This approach is based on authentic practices in paleontology and enables users to participate in activities that simulate real-world activities. The excavation process involves a series of interactions (e.g., tapping the dirt pile three times) which creates a sense of effort and advancement, and the cleaning process reveals the fossil bone fragment and leads the user to the subsequent interaction.

One of the critical design innovations of our system is the state-persistent interaction, which is not found in many current AR systems. Most AR systems have transient interactions; once users leave the application or walk away from the marker, their progress is lost. This breaks the learning flow and inhibits exploration. The system we're proposing preserves the state of each interaction using an ExhibitID-based state tracking mechanism, enabling users to pick up where they left off. For instance, if a user has cleaned the dirt off a fossil but leaves the game before answering a quiz, the system will retain this state and restore it when the user resumes the game.

This tracking feature is aligned with the principle of research-based learning, which is key to the system's design. Rather than requiring users to answer the quiz questions immediately, the system enables them to "soft exit" the quiz and obtain information from physical labels in the museum. This "soft exit" feature promotes active, environmental exploration, and helps integrate virtual and physical exploration. As a result, the system turns the museum experience from a passive to an active process of research.

The other key element of the system is the use of quiz-based learning as a progression mechanism. Rather than being an optional assessment, as is typically the case with quizzes, the quizzes in this system are gated. Players must successfully answer questions to earn fossil pieces and progress through the game. This approach ensures active recall, as users must recall and apply information to progress. Thus, learning is infused throughout the experience, rather than being a separate element.

The fossil fragments are stored in a virtual collection, which offers users a sense of accomplishment as they progress. When enough fragments are collected, users enter the skeleton assembly module to assemble the full skeleton of the prehistoric creature. At this level, users need to comprehend spatial awareness and structural accuracy to assemble the skeleton, which adds a greater degree of cognitive challenge.

The final step in the interaction sequence is the display of a fully assembled skeleton at full-scale (1:1), along with a virtual guide that offers information and "cool facts". This is a reward as well as an educational reinforcement mechanism, enabling the user to see the actual size and shape of the prehistoric animal - something that cannot be done with physical exhibits in Sri Lankan museums.

Besides interaction design, the system employs spatial control strategies to keep the user's attention and prevent tracking problems. A warning message is issued when the user moves beyond a certain threshold (e.g., 2 meters) and the session is paused when the user is too far away (e.g., 4 meters) with progress stored. This aspect avoids interaction failure caused by tracking loss and establishes the need to engage with the physical exhibit.

From an evaluation standpoint, the system provides support for quantitative analysis using telemetry. This approach differs from conventional museum studies which are based on observational techniques or questionnaires administered after the visit. These data allow for a more objective evaluation of user engagement and learning.

Moreover, this study employs an A/B testing approach, in which the engagement of users with the AR system is compared to the use of conventional labels. This enables the evaluation of the system's impact on engagement and learning, overcoming a major gap in previous research where such comparisons are typically missing.

Although AR technologies and research exist, few systems effectively combine interaction design, learning validation, state persistence and evaluation. Typically, these components are tackled separately, leading to disjointed experiences that don't fully harness the power of immersive technologies.

This study seeks to address these issues by proposing a system integrating:

- Structured AR interaction loops
- Gamified exploration mechanics
- Quiz-based knowledge validation
- Persistent user progression
- Data-driven evaluation

This research's emphasis on both educational and real-world system design considerations helps to inform the design of scalable, cost-effective and pedagogically sound AR systems for museums.

Although many recent studies have explored the use of Augmented Reality (AR) in educational and cultural settings, a more detailed analysis of current systems shows that most systems are still designed in an ad-hoc manner and lack pedagogic value. While individual elements like 3D visualisation, interaction or gamification may be present, these are not integrated into a unified and research-informed learning experience.

Most recent AR museum systems are centred on improving visualisation. They enable users to scan markers or surfaces to see 3D models of objects, animals or environments. Although this enhances user experience on a surface level, these systems tend to have shallow interaction design. Users are more passive and lack opportunities for decision-making, exploration and knowledge transfer. As a result, these platforms fail to work beyond visual enhancement to cognitive involvement, which is crucial for learning.

Furthermore, several educational AR systems have incorporated interactive features, such as quizzes, annotations, or tours. But these are often implemented as stand-alone modules, rather than part of a progression system. For instance, quizzes may be offered as optional activities that do not affect their experience. This allows users to skip these features without engaging in learning activities, limiting their effectiveness.

Another issue with existing systems is the lack of spatial interaction. Certain apps enable users to interact with digital objects in a way that is disconnected from the physical space, reducing the impact of the museum experience. However, our proposed system guarantees that users interact with a specific physical exhibit using image-based tracking. This keeps the digital and the physical world in context, enhancing learning.

Moreover, existing systems typically do not involve multi-step interaction processes that mimic real-world processes. Most interaction is confined to one-step processes (such as scan and view) without evolving or changing. This is in contrast to the design of this research, where interaction takes place in a multi-stage process (excavate → clean → quiz → unlock → collect). This not only keeps users engaged, but it also simulates actual scientific workflows, contributing to authenticity and learning outcomes.

A major weakness of current systems is the absence of persisting state and continuity. Typical AR systems have ephemeral user interactions. When the app is terminated or the marker is removed, the system is restarted and the user's state is lost. This approach doesn't align with typical museum interactions, where users can navigate between exhibits, stop and start interactions, or return to previous topics. The non-persistent nature of the system affects learning and drives away users. The interaction system in this research overcomes this problem by having a state-aware system, in which each interaction is saved and restored according to an exhibit identifier.

Another key issue is the absence of learning mechanisms. Although some systems provide information, this is not verified. This leads to passive learning rather than active participation. By contrast, the developed system includes gating through quizzes where users must answer questions correctly to progress. This promotes active learning and retrieval-based learning strategies.

In terms of evaluation, many current AR systems do not include formal and measurable evaluation techniques. The positive effects on engagement or learning are often subjective user feedback or observational studies, which are not precise and scalable.

There are few systems that incorporate structured experimentation like controlled A/B testing, which allows for outcomes to be compared with a control situation. This lack of empirical evidence is a major bottleneck.

The system proposed in this paper overcomes this by including:

- Objective measures (e.g., time on task, completion, quiz scores)
- Controlled experiment design (AR-based vs. label-based learning)

This allows for a more comprehensive assessment of the system, with empirical data to support its value.

Technically, current AR systems typically use either QR code-based tracking or markerless tracking. QR codes are efficient and robust, but lack flexibility and may be visually intrusive in museum settings. Markerless tracking offers more flexibility but is prone to issues of performance, stability and environmental factors.

This research project has developed a hybrid system using custom-designed image markers ("White X") that combine stability with aesthetics. This marker is designed to provide good feature recognition but to be unobtrusive in the exhibition environment. This ensures reliable tracking, without compromising the exhibit's aesthetic.

A further novel aspect of the proposed system is the use of distance-based interaction control. Most AR systems do not consider the user's distance to the target, which may result in poor tracking and interaction. Through the inclusion of distance thresholds (e.g., warnings at 2 meters and session termination at 4 meters), the system ensures the best interaction and keeps the progress.

Although there are numerous AR applications, it is clear that most systems focus on separated parts of interaction, learning or assessment, rather than a holistic approach. This hampers their effectiveness and their use in educational practice.

On the other hand, the system presented in this study is an integrated learning environment in which:

- Interaction is structured and sequential
- A learning quiz is required
- State is preserved for progress
- Learning is tracked through telemetry and experimentation

Through the integration of these elements, this research goes beyond the implementation of individual features, and provides a comprehensive approach to AR-based museum learning.

1.2 Research Problem

The primary research question addressed in this study is:

How can a phone-based augmented reality (AR) system integrated with quiz-driven interactions and state-persistent learning mechanisms be designed to transform static fossil exhibits into interactive learning experiences that enhance user engagement and knowledge retention?

Traditional museum environments, particularly in Sri Lanka, rely heavily on static displays and textual labels, which often result in passive learning experiences and limited user engagement. As highlighted in the project proposal, fossil exhibits lack interactive elements that can effectively capture the attention of younger audiences and encourage meaningful exploration . Visitors typically observe exhibits briefly and move on without deep cognitive involvement, leading to low retention of educational content.

To address this limitation, this research investigates the design and implementation of an AR-based interactive system that combines real-world exploration with digital learning elements. The proposed system introduces a structured interaction loop involving fossil discovery, excavation, quiz-based validation, and skeleton reconstruction, thereby transforming passive observation into an active learning process.

This research problem is inherently multidimensional and involves several key challenges:

Key Challenges

- **Designing an interactive AR-based learning workflow**
  Developing a seamless interaction flow that integrates physical exhibit exploration with digital AR elements, ensuring intuitive user experience and engagement.

- **Replacing passive learning with active knowledge acquisition**
  Encouraging users to actively search for information, answer quiz questions, and engage in problem-solving rather than relying on passive reading of labels.

- **Implementing state-persistent interaction mechanisms**
  Ensuring that user progress, including excavation and quiz states, is preserved across sessions, allowing non-linear exploration consistent with real museum behavior.

- **Balancing engagement with educational effectiveness**
  Designing gamified elements such as scavenger hunts, rewards, and interactive tools without compromising the educational value of the content.

- **Maintaining real-time performance on mobile devices**
  Ensuring stable AR tracking, responsive interactions, and efficient rendering using smartphone hardware within a museum environment.

- **Evaluating user engagement and learning outcomes**
  Measuring the effectiveness of the system using quantitative metrics such as interaction time, completion rates, and quiz performance, alongside comparative analysis with traditional methods.

1.3 Research Gap

While the integration of Augmented Reality (AR) into educational and museum environments has gained significant attention in recent years, a closer examination of existing systems reveals that most implementations remain fragmented in design and limited in pedagogical effectiveness. Although individual components such as 3D visualization, interaction, or gamification are often present, they are rarely combined into a cohesive and research-driven learning framework.

Many contemporary AR-based museum applications primarily focus on enhancing visual representation. These systems allow users to scan markers or surfaces to view reconstructed 3D models of artifacts, animals, or environments. While this improves user engagement at a superficial level, such approaches often lack depth in interaction design. The user’s role remains largely observational, with limited opportunities for decision-making, exploration, or knowledge application. Consequently, these systems fail to transition from visual augmentation to cognitive engagement, which is essential for meaningful learning.

In addition, a number of educational AR platforms have introduced interactive elements such as quizzes, annotations, or guided tours. However, these features are frequently implemented as independent modules, rather than being integrated into a unified progression system. For example, quizzes may be presented as optional activities that do not influence the user’s overall experience. As a result, users can bypass these elements without engaging in active learning, reducing their educational impact.

Another limitation observed in existing systems is the absence of contextual interaction tied to physical space. Some applications allow users to interact with virtual objects independently of their physical surroundings, which diminishes the relevance of the museum environment itself. In contrast, the system proposed in this research ensures that all interactions are anchored to specific physical exhibits through image-based tracking. This maintains a strong connection between digital content and real-world artifacts, reinforcing contextual learning.

Furthermore, current implementations rarely incorporate multi-stage interaction pipelines that simulate real-world processes. Most interactions are limited to single-step actions, such as scanning and viewing, without progression or transformation. This contrasts with the approach adopted in this research, where interaction is structured as a sequential process (excavation → cleaning → quiz → unlocking → collection). This design not only increases engagement but also mirrors real-world scientific practices, thereby enhancing authenticity and educational value.

A critical gap in existing systems is the lack of state persistence and continuity. In most AR applications, user interactions are temporary and session-based. Once the application is closed or the marker is lost, the system resets, and all progress is lost. This design is incompatible with real-world museum behavior, where visitors may move between exhibits, pause interactions, or revisit content. The absence of persistence disrupts the learning process and discourages extended engagement. The system developed in this research addresses this issue through a state-aware architecture, where each interaction is saved and restored based on a unique exhibit identifier.

Another important limitation relates to the lack of enforced learning mechanisms. While some systems include informational content, they do not require users to demonstrate understanding. This results in passive consumption of information rather than active engagement. In contrast, the proposed system integrates quiz-based gating, where progression is contingent upon correct answers. This ensures that users actively engage with the content and reinforces learning through retrieval practice.

From an evaluation standpoint, many existing AR applications lack rigorous and quantifiable assessment methods. Claims regarding improved engagement or learning are often based on subjective feedback or observational studies, which are limited in accuracy and scalability. Few systems implement structured experimental designs such as controlled A/B testing, where outcomes can be directly compared against a baseline condition. This lack of empirical validation represents a significant limitation in current research.

The system presented in this study addresses this issue by incorporating both:

- Quantitative metrics (e.g., dwell time, completion rates, quiz accuracy)
- Controlled experimental comparison (AR-based interaction vs. label-based learning)

This enables a more robust evaluation of the system's effectiveness, providing measurable evidence of its impact.

In terms of technical implementation, existing AR systems often rely on either QR code-based tracking or markerless tracking approaches. QR codes are simple and reliable but offer limited flexibility and can disrupt the visual aesthetics of museum environments. Markerless tracking, on the other hand, provides greater flexibility but introduces challenges related to performance, stability, and environmental dependency.

The system developed in this research adopts a hybrid approach using custom-designed image targets ("White X"), which balance reliability and visual integration. These markers are specifically designed to provide strong feature detection while remaining minimally intrusive within the exhibit space. This approach ensures consistent tracking performance while maintaining the visual integrity of the museum environment.

Another distinguishing feature of the proposed system is the incorporation of proximity-aware interaction control. Many existing AR applications do not account for the user's spatial relationship to the target, which can lead to unstable tracking and inconsistent interaction. By introducing distance thresholds (e.g., warnings at 2 meters and session termination at 4 meters), the system maintains optimal interaction conditions and preserves user progress.

Despite the wide range of AR applications available, it is evident that most systems address only individual aspects of interaction, learning, or evaluation, rather than integrating them into a unified framework. This fragmentation limits their effectiveness and reduces their applicability in real-world educational settings.

In contrast, the system proposed in this research is designed as an integrated learning ecosystem, where:

- Interaction is structured and sequential
- Learning is enforced through quiz-based validation
- Progress is maintained through state persistence
- Engagement is measured through telemetry and experimentation

By synthesizing these components into a single system, this research moves beyond isolated feature implementation and contributes a holistic approach to AR-based museum learning.

1.4 Research Problem

Although there is increasing interest in immersive technologies in education and cultural heritage, fossil displays in Sri Lankan museums are still largely based on traditional static display techniques that hinder learning and engagement. Such exhibits typically feature a fossil, skeleton, or written descriptions, which requires users to infer and recreate the ecological environment and lifestyles of extinct organisms. Although scientifically correct, this approach is cognitively demanding for users, especially children, leading to a passive viewing experience and low retention rates.

This is because current museum exhibits do not contain any interactive and structured learning processes. Users have a linear, observational approach to viewing exhibits, with no learning experience to guide them. No feedback mechanisms are provided to stimulate discovery, assess comprehension or reinforce learning. Thus, museum experiences are not engaging or cognitively stimulating.

Many have suggested using Augmented Reality (AR) systems to improve museum experiences by adding digital information to exhibits. But existing AR systems do not completely tackle the drawbacks of these conventional systems. Most applications are focused on the visual improvement, by providing a 3D model, or animation, when the user is scanning a marker. These enhancements help to increase engagement on a surface level, but they don't necessarily encourage understanding and learning. As a result, AR tends to be used for visualisation rather than learning.

Lack of interaction workflows is another area of concern. Current systems usually offer discrete interactions, such as scanning and viewing, but don't integrate these interactions into a progression model. This leads to disjointed user experiences with isolated interactions. Users are not led through a process of activities that build on the previous ones, thus preventing meaningful learning and discovery.

Additionally, current systems do not typically include active learning strategies, such as quizzing, as an integral part of the interactive design. Although some applications have informational content, they seldom require users to demonstrate knowledge of the content.

This results in passive information processing, which has been demonstrated to be less effective for long-term retention than other methods based on recall.

Another problem is the non-persistent nature of user interaction. Existing AR systems are often session-based - once the user leaves the application or moves out of marker view, their progression is lost. This is not an effective approach based on typical museum use, which often involves moving around the museum, pausing interactions or revisiting information. A lack of persistence negatively impacts learning, interaction and engagement.

Beyond the limitations in interaction and learning, existing systems also lack rigorous and empirical evaluation methods. A range of AR museum experiences are claimed to benefit engagement and learning effectiveness; however, they lack empirical validation. Without rigorous evaluation strategies (such as A/B testing), it is not possible to ascertain whether immersive technologies genuinely improve learning outcomes, by offering only short-term engagement.

In terms of implementation, many sophisticated AR applications use complex or resource-heavy techniques, such as markerless tracking or dedicated hardware. These approaches can cause performance, stability and cost issues, and are therefore not appropriate for Sri Lankan museums. We need a practical, affordable and scalable approach that works on off-the-shelf mobile devices, with stable performance.

These challenges demonstrate the presence of a problem:

Current museum offerings, as well as existing AR solutions, do not offer an integrated, interactive and measurable experience that enhances user engagement and learning retention.

To solve this problem, it is not enough to simply add a visual element; a system is needed to build a holistic learning experience that incorporates interaction, active learning, persistence and assessment. This system must lead the user through a learning process by guiding them, promoting discovery, confirming their knowledge, and measuring their learning experiences.

To this end, the investigation is guided by the research question:

How would a mobile augmented reality system, which includes interaction workflows, state persistence and quiz-based skill validation, make fossil displays interactive and enhance user engagement and learning outcomes in comparison to label-based approaches?

These question will guide the development and evaluation of the proposed system, which will transcend the conventional and current approaches to interactive museum learning, and provide a scalable and empirically proven framework for interactive learning in museums.

1.5 Research Objectives

Main objective

The primary objective of this research is to design, develop, and evaluate a smartphone-based augmented reality (AR) learning system that transforms fossil exhibits into interactive and structured learning experiences. The system aims to improve user engagement and knowledge retention by integrating gamified interaction workflows, state persistence, and quiz-based knowledge validation, while ensuring practical deployment within real-world museum environments.

Specific objectives

To effectively attain the overall objective in a systematic manner, the research is informed by the following detailed and interrelated specific objectives:

1. To design and implement a **marker-based AR system** using image tracking to achieve stable and accurate alignment of virtual content with physical museum exhibits.
2. To develop a **sequential interaction model** incorporating excavation, cleaning, quiz-based unlocking, and collection processes to guide users through a structured learning workflow.
3. To integrate **gamification techniques**, including guided navigation, tool-based interactions, and reward-driven progression, to enhance user motivation and exploration.
4. To implement a **state persistence mechanism** that maintains user progress across sessions by tracking interaction states associated with individual exhibit identifiers.
5. To incorporate **quiz-based active recall mechanisms** as a mandatory component of progression, ensuring that users engage cognitively with exhibit-related information.

6. To design and develop a **virtual inventory and skeleton assembly module** that enables users to reconstruct complete fossil structures, reinforcing spatial understanding and learning outcomes.
7. To implement **proximity-aware interaction control** to ensure stable AR tracking and maintain user focus within optimal interaction distances.
8. To capture and analyze **user interaction data through telemetry**, including dwell time, completion rates, and interaction patterns, for objective evaluation of engagement.
9. To conduct a **controlled A/B evaluation**, comparing the AR-based system with traditional label-based learning, in order to measure improvements in engagement and knowledge retention.
10. To evaluate the feasibility of deploying a **low-cost, scalable AR solution** suitable for Sri Lankan museum environments using standard mobile devices.

2. Methodology

2.1 Methodology Overview

The *Fossil Hunt AR* system is a mobile-based augmented reality learning application designed to enhance the educational experience of museum visitors by transforming traditional fossil exhibits into interactive, engaging, and structured learning environments. The system integrates physical exploration with digital interaction, enabling users to actively participate in a guided learning process rather than passively observing static displays.

At its core, the system is built around the concept of a gamified research experience, where users assume the role of an explorer tasked with discovering and reconstructing fossil fragments distributed throughout the museum space. This approach shifts the user's role from that of a passive observer to an active participant, encouraging curiosity, exploration, and engagement with both the physical and digital aspects of the environment.

The system operates through a sequential interaction workflow, where each stage represents a distinct phase of user engagement. Unlike conventional AR applications that provide isolated interactions, the proposed system establishes a continuous and structured progression that guides users through multiple layers of activity. This progression is carefully designed to balance physical interaction, cognitive engagement, and reward-based motivation, ensuring that learning is embedded within the user experience.

The interaction begins with a guided introduction, where users are provided with instructions and contextual information through an in-application interface. This stage introduces the concept of locating specific markers within the environment, which serve as entry points for AR interactions. The use of a treasure-map metaphor plays a crucial role in framing the experience as an exploratory mission, thereby increasing user motivation and engagement.

Once users begin exploring the environment, they encounter designated image-based markers placed near physical exhibits. These markers act as triggers for augmented reality content, enabling the system to overlay digital objects onto real-world surfaces. This integration ensures that the AR experience remains contextually tied to the museum environment, reinforcing the connection between physical artifacts and digital enhancements.

Upon detecting a marker, the system initiates an interactive excavation process, where users are required to perform specific actions using virtual tools. This stage introduces a layer of experiential interaction, simulating real-world fossil excavation practices. The use of tools such as a shovel and brush allows users to engage in meaningful actions that mirror actual scientific procedures, thereby enhancing the authenticity of the experience.

Following the excavation phase, users transition to a cleaning stage, where the fossil is revealed and refined through additional interaction. This progression from excavation to cleaning represents a deliberate design choice, as it introduces a sense of continuity and gradual discovery. By structuring the interaction in stages, the system maintains user engagement and reinforces the concept of progression.

A key component of the system is the integration of quiz-based knowledge validation, which serves as the primary mechanism for reinforcing learning. After completing the physical interaction stages, users are presented with a question related to the exhibit. This requires them to recall and apply information, shifting the focus from physical interaction to cognitive engagement. Unlike traditional systems where information is passively consumed, this approach ensures that users actively process and understand the content.

An important feature of the system is its support for non-linear learning behavior, achieved through a "soft exit" mechanism. If users are unable to answer a question, they are allowed to exit the quiz, explore the surrounding environment, and return later to complete the task. This design encourages users to engage with real-world exhibit information, effectively bridging the gap between digital interaction and physical learning.

Upon successfully answering the quiz, users are rewarded with a fossil fragment, which is added to their virtual collection. This reward system introduces a sense of achievement and provides immediate feedback, motivating users to continue their exploration. The system maintains a visible record of collected fragments, allowing users to track their progress and understand how close they are to completing the overall objective.

The final stage of the system involves the assembly and visualization of a complete fossil structure. Once all fragments have been collected, users can view a full-scale augmented reality representation of the reconstructed skeleton. This stage serves as both a reward and an educational tool, allowing users to perceive the true scale and structure of the organism, which is often not possible through physical exhibits alone.

From a technical perspective, the system is designed to operate efficiently on standard mobile devices, ensuring accessibility and scalability. The use of image-based tracking provides reliable performance in indoor environments, while the modular architecture allows for easy extension and adaptation to different exhibits.

Overall, the *Fossil Hunt AR* system represents a shift from traditional, static learning methods to a dynamic, interactive, and research-driven approach. By integrating exploration, interaction, knowledge validation, and reward into a unified framework, the system provides a comprehensive solution to the limitations identified in existing museum experiences. The following sections describe the system design, architecture, and implementation in greater detail.

2.2 Guided Introduction and Exploration Mechanism

The interaction begins with a **guided tutorial interface**, which serves as the entry point to the system. At this stage, users are introduced to the objectives of the application through a visual and narrative-based instruction system. A virtual character provides guidance, explaining that specific locations within the environment are marked and must be discovered in order to proceed.

Figure 1

The use of a treasure-map metaphor plays a significant role in shaping user perception. By framing the experience as an exploratory mission, the system encourages users to actively search for interaction points rather than passively observe their surroundings. This approach leverages gamification principles to increase motivation and curiosity, particularly among younger users.

From a methodological perspective, this stage is essential for reducing initial user uncertainty and establishing a clear sense of direction. It ensures that users understand both the purpose of the system and the steps required to achieve their objectives, thereby facilitating a smooth transition into the interactive phases.

2.3 Tool-Based Interaction Framework

Following the introduction, users are presented with a tool-based interaction interface, which forms the foundation of the system's experiential design. The primary tools available to the user are a shovel and a brush, each associated with a specific stage of the interaction process.

Figure 2

The inclusion of tools is a deliberate design decision aimed at simulating real-world fossil excavation practices. By requiring users to select and use appropriate tools, the system introduces an element of task-based interaction, where actions are purposeful and context-dependent. This enhances the realism of the experience and strengthens the connection between the digital system and real-world scientific processes.

The tool-based framework also introduces interaction variability, as users must switch between tools depending on the stage of the process. This prevents repetitive interaction patterns and maintains user engagement by requiring continuous attention and decision-making.

2.3 AR Marker Detection and Excavation Process

The core augmented reality functionality is activated when the user identifies and scans a designated image marker, represented as a “White X” within the environment. These markers are strategically placed near physical exhibits, ensuring that the AR interaction remains contextually relevant.

Figure 3

Upon successful detection of the marker, the system generates a three-dimensional dirt pile superimposed onto the real-world surface. This marks the beginning of the excavation phase, where users must interact with the dirt using the shovel tool.

The excavation process is designed to require multiple user inputs, typically in the form of repeated taps. This design introduces a sense of effort and progression, as the dirt is gradually removed rather than instantly disappearing. By requiring sustained interaction, the system encourages users to remain engaged and actively participate in the process.

This stage establishes a critical link between physical space and digital interaction, reinforcing the concept that learning is tied to real-world context. It also serves as the foundation for subsequent stages, as the completion of excavation is necessary to reveal the next layer of interaction.

2.4 Cleaning and Fossil Reveal Mechanism

Following the excavation phase, the system transitions into the **cleaning stage**, where the fossil is revealed in a partially obscured state. At this point, users must switch to the brush tool to remove the remaining dirt and fully uncover the fossil.

Figure 4

This stage introduces a more refined form of interaction, requiring precision rather than repetition. The transition from excavation to cleaning mirrors real-world processes, enhancing the authenticity of the experience and reinforcing the educational context.

The visual transformation of the fossil—from a dirt-covered object to a clean and recognizable structure—provides immediate feedback to the user. This reinforces the sense of accomplishment and maintains engagement as users progress toward the next stage.

2.5 Quiz-Based Knowledge Validation Mechanism

Once the fossil has been fully revealed, the system introduces a quiz-based interaction, which serves as the primary mechanism for knowledge validation. This stage represents a shift from physical interaction to cognitive engagement, requiring users to recall and apply information related to the exhibit.

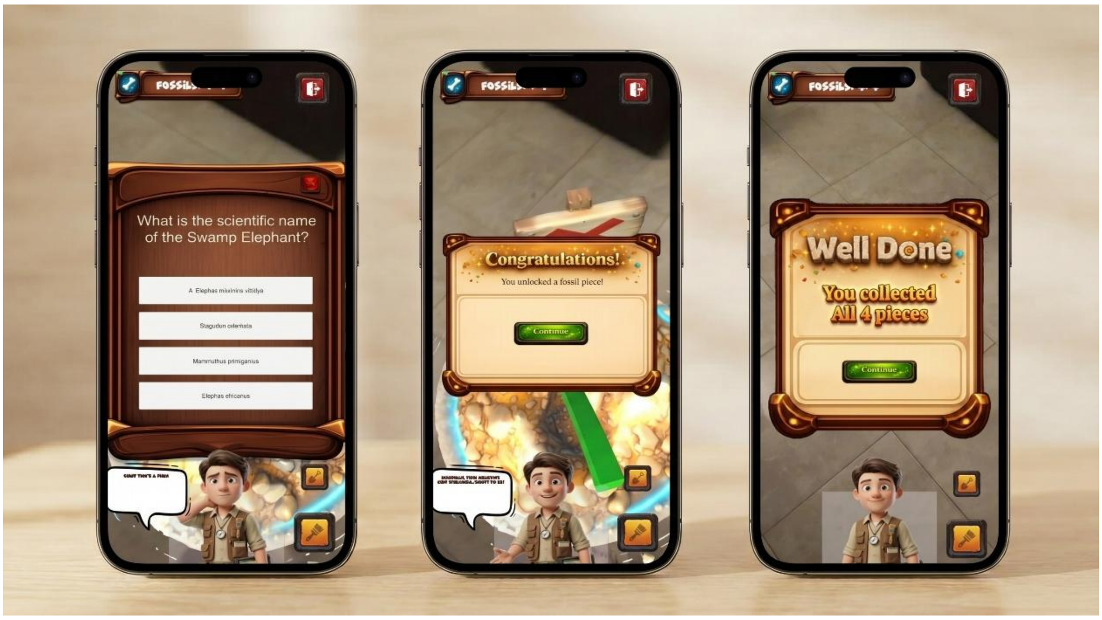


Figure 5

The quiz is presented in a multiple-choice format, allowing users to select from predefined answers. However, the system incorporates a unique feature known as the "soft exit" mechanism, which allows users to temporarily exit the quiz if they are unsure of the answer. This encourages them to explore the surrounding environment, read informational labels, and return with the necessary knowledge to complete the task.

This design effectively bridges the gap between digital interaction and physical learning, transforming the museum visit into an active research process. Rather than relying solely on the application for information, users are encouraged to engage with the real-world exhibits, thereby enhancing the overall learning experience.

2.6 Reward System and Progression Tracking

Upon successfully answering the quiz, users are rewarded with a **fossil fragment**, which is added to their virtual collection. This reward system plays a crucial role in maintaining user motivation and reinforcing positive behavior.

The system provides immediate feedback by displaying the number of collected fragments relative to the total required. This creates a clear sense of progression and encourages users to continue exploring additional markers within the environment.

The reward mechanism is closely tied to the concept of **goal-oriented interaction**, where each completed task contributes to a larger objective. This not only enhances engagement but also provides a structured framework for the learning process.

2.7 Final Assembly and AR Visualization

After collecting all required fragments, users unlock the final stage of the interaction flow, which involves the assembly and visualization of a complete fossil structure.

Figure 6

In this stage, the system presents a full-scale (1:1) augmented reality model of the reconstructed skeleton, allowing users to observe its size and structure within their real-world environment. This provides a level of visualization that is not achievable through traditional museum displays.

The inclusion of a virtual guide that delivers additional information further enhances the educational value of this stage, reinforcing the knowledge acquired throughout the interaction process.

2.8 Summary of Interaction Design

The interaction flow of the *Fossil Hunt AR* system demonstrates a carefully structured progression from exploration to interaction and finally to knowledge validation. By integrating multiple layers of engagement, the system ensures that users remain actively involved throughout the experience.

The combination of gamification, experiential interaction, and cognitive reinforcement creates a comprehensive learning environment that addresses the limitations of traditional museum exhibits. Each stage of the interaction flow contributes to a cohesive and meaningful user experience, ultimately supporting the research objective of enhancing engagement and knowledge retention.

2.9 System Architecture

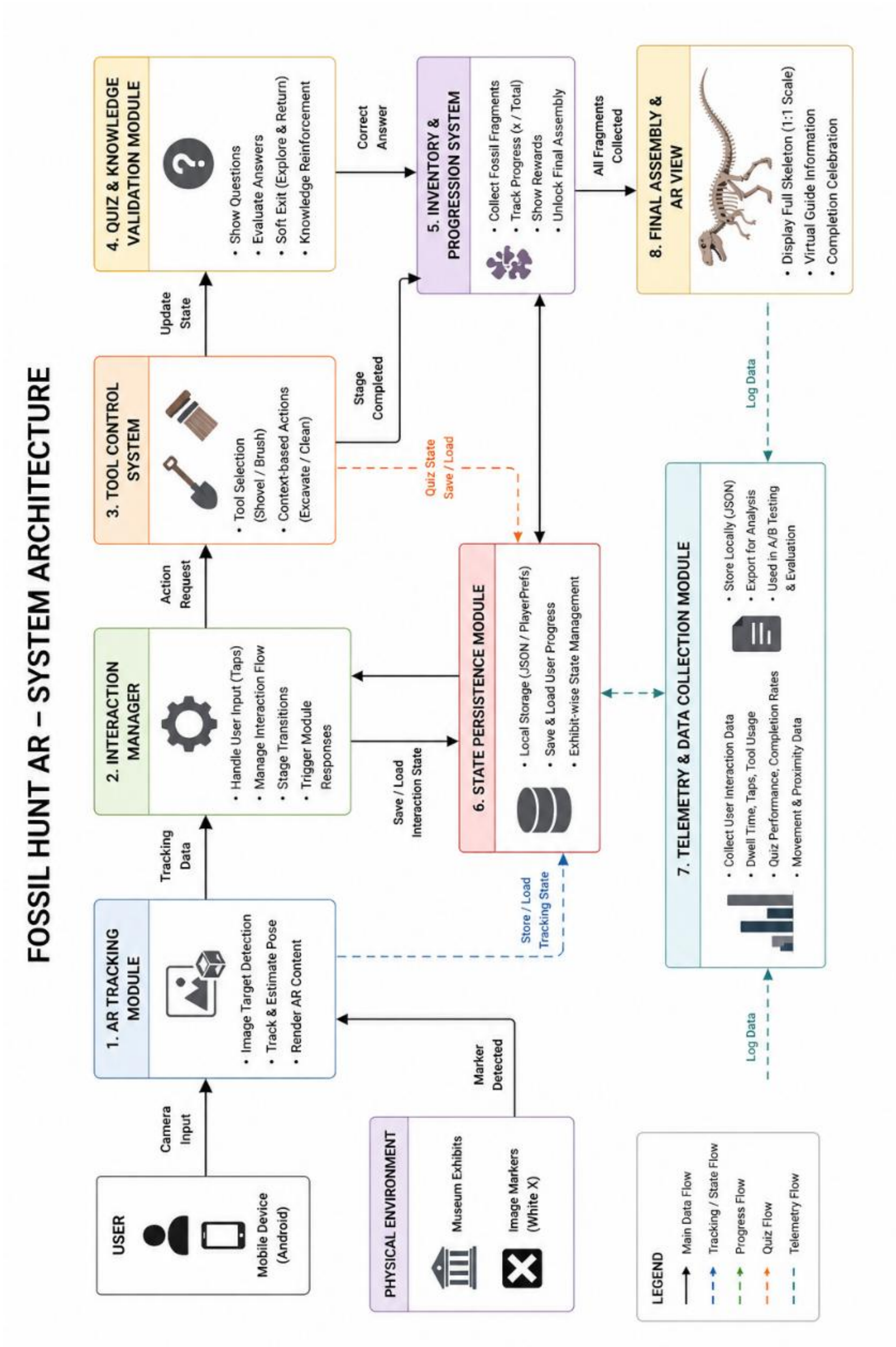
The architecture of the *Fossil Hunt AR* system is designed to support a modular, scalable, and real-time interactive augmented reality experience within a mobile environment. The system integrates multiple functional components that collectively enable marker detection, user interaction, learning validation, and data tracking. This architecture is implemented using Unity 6 as the core development engine and the Vuforia Engine for image-based augmented reality tracking, forming a robust foundation for building interactive AR applications.

At a high level, the system follows a layered architecture, where each layer is responsible for a specific set of functionalities. This modular design ensures that individual components can be developed, tested, and maintained independently, while still functioning cohesively as part of the overall system.

2.9.1 Architecture Overview

The system architecture consists of several key modules, each responsible for handling a specific aspect of the application:

- AR Tracking Module
- Interaction Manager
- Tool Control System
- Quiz and Knowledge Validation Module
- State Persistence Module
- Inventory and Progression System
- Telemetry and Data Collection Module



FOSSIL HUNT AR – SYSTEM ARCHITECTURE
USER
Mobile Device (Android)
Camera Input
1. AR TRACKING MODULE
• Image Target Detection
• Track & Estimate Pose
• Render AR Content
Tracking Data
2. INTERACTION MANAGER
• Handle User Input (Taps)
• Manage Interaction Flow
• Stage Transitions
• Trigger Module Responses
Action Request
3. TOOL CONTROL SYSTEM
• Tool Selection (Shovel / Brush)
• Context-based Actions (Excavate / Clean)
Update State
4. QUIZ & KNOWLEDGE VALIDATION MODULE
• Show Questions
• Evaluate Answers
• Soft Exit (Explore & Return)
• Knowledge Reinforcement
Correct Answer
Stage Completed
5. INVENTORY & PROGRESSION SYSTEM
• Collect Fossil Fragments
• Track Progress (x / Total)
• Show Rewards
• Unlock Final Assembly
All Fragments Collected
8. FINAL ASSEMBLY & AR VIEW
• Display Full Skeleton (1:1 Scale)
• Virtual Guide Information
• Completion Celebration
PHYSICAL ENVIRONMENT
Museum Exhibits
Image Markers (White X)
Marker Detected
Store / Load Tracking State
Save / Load Interaction State
Quiz State Save / Load
6. STATE PERSISTENCE MODULE
• Local Storage (JSON / PlayerPrefs)
• Save & Load User Progress
• Exhibit-wise State Management
Log Data
Log Data
7. TELEMETRY & DATA COLLECTION MODULE
• Collect User Interaction Data
• Dwell Time, Taps, Tool Usage
• Quiz Performance, Completion Rates
• Movement & Proximity Data
• Store Locally (JSON)
• Export for Analysis
• Used in A/B Testing & Evaluation
LEGEND
Main Data Flow
Tracking / State Flow
Progress Flow
Quiz Flow
Telemetry Flow

Figure 7

These modules interact through a centralized control flow, ensuring that user actions trigger appropriate system responses while maintaining synchronization between physical inputs and digital outputs.

2.10 AR Tracking Module

The AR Tracking Module is responsible for detecting and tracking image-based markers within the environment. The system utilizes Vuforia's Image Target technology, which allows predefined visual markers (White X indicators) to be recognized through the device's camera.

When a marker is detected:

- The camera feed is processed in real time
- The corresponding image target is identified
- A virtual object (e.g., dirt pile) is instantiated at the marker location

This module ensures that digital content is accurately aligned with the physical environment, providing a seamless augmented reality experience. The use of image-based tracking is particularly advantageous in indoor museum settings, as it offers high stability and precision compared to markerless approaches, which may be affected by lighting conditions and surface features.

2.11 Interaction Manager

The Interaction Manager acts as the **central control unit** of the system, coordinating user input, system states, and interaction flow. It is responsible for:

- Handling user touch input (tap interactions)
- Managing transitions between interaction stages
- Triggering appropriate system responses based on user actions

For example, when a user taps on the dirt pile during the excavation phase, the Interaction Manager processes the input and updates the excavation state. Once the required number of interactions is completed, it triggers the transition to the cleaning phase.

This centralized control mechanism ensures that the system maintains a consistent and structured interaction flow, preventing conflicts between different modules.

2.12 Tool Control System.

The Tool Control System manages the selection and usage of interaction tools, specifically the shovel and brush. This module is closely integrated with the Interaction Manager and determines how user inputs are interpreted based on the selected tool.

- When the shovel is active, input is interpreted as excavation
- When the brush is active, input is interpreted as cleaning

The system ensures that only contextually appropriate actions are executed, preventing unintended interactions. This design enhances usability by providing clear feedback and maintaining logical consistency within the interaction flow.

2.13 Quiz and Knowledge Validation Module

The Quiz Module is responsible for presenting questions, processing user responses, and determining progression based on answer correctness. This module is activated after the cleaning phase and serves as the primary mechanism for learning validation.

Key functionalities include:

- Displaying multiple-choice questions
- Evaluating selected answers
- Triggering rewards or retry mechanisms

The module is also integrated with the soft exit feature, allowing users to temporarily exit the quiz and return later without losing progress. This requires coordination with the State Persistence Module to store the current quiz state.

2.14 State Persistence Module

The State Persistence Module is a critical component of the system, responsible for maintaining user progress across sessions. Each interaction is associated with a unique ExhibitID, which allows the system to track the state of individual exhibits.

The module stores information such as:

- Whether excavation has been completed
- Whether the fossil has been cleaned
- Quiz progress and completion status

```
2 references
public void DisplayStep() {
    if (currentStep >= tutorialSteps.Count) return;

    TutorialStep step = tutorialSteps[currentStep];

    // 1. --- AUDIO HANDLING ---
    if (voiceSource != null && step.voiceClip != null) {
        voiceSource.Stop(); // Reset any playing audio
        voiceSource.clip = step.voiceClip;
        voiceSource.Play(); // Play current track
    }

    // 2. --- COROUTINE MANAGEMENT ---
    // We stop all previous typing and scaling animations before starting new ones
    StopAllCoroutines();
    StartCoroutine(TypeText(step.text));

    // Start timed sprite swaps for this track
    foreach (var change in step.spriteSequence) {
        StartCoroutine(WaitAndSwapSprite(change.pose, change.timeToSwap));
    }

    // 3. --- ICON VISIBILITY & SCALING ---
    if (shovelIcon != null) {
        shovelIcon.SetActive(step.showShovel);
        float sTarget = step.showShovel ? zoomedScale : 1.0f;
        StartCoroutine(ScaleIcon(shovelRect, sTarget));
    }

    if (brushIcon != null) {
        brushIcon.SetActive(step.showBrush);
        float bTarget = step.showBrush ? zoomedScale : 1.0f;
        StartCoroutine(ScaleIcon(brushRect, bTarget));
    }
```

```
    if (trackerIcon != null) trackerIcon.SetActive(step.showTracker);

    // 4. --- FINAL STEP TRANSITION ---
    if (step.isFinalStep) {
        if(nextButton != null) nextButton.SetActive(false);
        if(readyButton != null) readyButton.SetActive(true);
    }
}
```

Figure 8

When a user revisits an exhibit, the system retrieves the stored state and restores the interaction to its previous stage. This ensures continuity and supports non-linear exploration, allowing users to move freely between exhibits without losing progress.

2.15 Inventory and Progression System

The Inventory System manages the collection of fossil fragments and tracks user progression throughout the experience. Each successfully completed interaction results in the addition of a fragment to the user's inventory.

The system maintains:

- A count of collected fragments
- The total number required for completion
- Progress indicators displayed to the user

This module is closely linked to the reward system, providing feedback and motivating users to continue interacting with the application.

2.16 Telemetry and Data Collection Module

The Telemetry Module is responsible for capturing user interaction data, which is used for evaluation and analysis. This includes:

- Time spent on each interaction (dwell time)
- Number of interactions performed
- Quiz performance and accuracy
- Movement patterns and proximity data

The collected data is used to support the evaluation phase of the research, enabling a quantitative analysis of user engagement and learning outcomes.

2.17 Data Flow and System Integration

The overall system operates through a continuous data flow process:

1. The AR Tracking Module detects a marker
2. The Interaction Manager initializes the interaction
3. User input is processed through the Tool Control System
4. Interaction state is updated and stored in the State Persistence Module
5. The Quiz Module validates knowledge
6. The Inventory System updates progress
7. The Telemetry Module records interaction data

This integrated workflow ensures that all system components operate cohesively, providing a seamless and responsive user experience.

2.18 Summary of System Architecture

The architecture of the *Fossil Hunt AR* system is designed to support a real-time, interactive, and data-driven learning environment. By adopting a modular approach, the system ensures flexibility, scalability, and maintainability, while also enabling the integration of multiple functionalities such as AR tracking, interaction management, and data analysis.

The combination of robust tracking, structured interaction control, persistent state management, and telemetry-based evaluation provides a comprehensive framework for delivering an effective augmented reality learning experience. This architectural design directly supports the research objectives by enabling both interactive engagement and measurable outcomes, forming the foundation for the system's implementation and evaluation.

2.19 State Persistence Mechanism

The *Fossil Hunt AR* system incorporates a state persistence mechanism as a core architectural component to support continuous, non-linear, and user-centered interaction within the museum environment. Unlike conventional augmented reality applications that operate on a session-based model, where user progress is lost once the interaction ends, this system is designed to retain and restore interaction states across multiple sessions and physical locations. This feature addresses a critical limitation in existing AR systems and aligns with real-world user behavior in museum contexts.

In a typical museum setting, users do not interact with exhibits in a strictly linear manner. Visitors often move freely between different exhibits, pause interactions, or revisit previously explored sections. Therefore, a system that requires users to restart interactions from the beginning each time would disrupt the learning experience and reduce engagement. The state persistence mechanism is implemented to overcome this limitation by ensuring that user progress is preserved and can be resumed seamlessly at any point.

Conceptual Design of State Persistence

The state persistence mechanism is designed around the concept of exhibit-specific state tracking, where each interaction point within the system is uniquely identified and managed independently. Each AR marker (White X) is associated with a unique identifier, referred to as the ExhibitID. This identifier serves as the key for storing and retrieving the state of a particular interaction.

For each ExhibitID, the system maintains multiple state variables, including:

- **Excavation State** – Indicates whether the dirt pile has been cleared
- **Cleaning State** – Indicates whether the fossil has been cleaned
- **Quiz State** – Tracks whether the quiz has been completed or is in progress
- **Completion State** – Indicates whether the fragment has been successfully collected

By structuring the system in this way, each exhibit functions as an independent unit within the overall interaction framework. This allows users to interact with exhibits in any order while maintaining consistency and continuity.

2.20 State Storage and Data Management

The system stores user progress locally on the device using lightweight data storage mechanisms, such as JSON-based storage or Unity's PlayerPrefs system. This approach ensures that the persistence mechanism remains efficient and does not require external servers or network connectivity, making it suitable for real-world deployment in museum environments.

2.21 State Restoration and Interaction Continuity

When a user scans an AR marker, the system first queries the State Persistence Module to determine the current state associated with the corresponding ExhibitID. Based on the retrieved data, the system dynamically adjusts the interaction flow.

For example:

- If the excavation state is already completed, the system skips the excavation phase and directly displays the cleaned or partially cleaned fossil
- If the quiz has not been completed, the system resumes the quiz interface from its previous state
- If the fragment has already been collected, the system prevents redundant interaction and provides appropriate feedback

This dynamic behavior ensures that the user experience remains consistent and uninterrupted, regardless of when or how the user returns to a particular fossilpart.

2.22 Soft Exit and Resume Mechanism

A key feature enabled by the state persistence system is the implementation of a soft exit mechanism during the quiz phase. If a user is unable to answer a question, they can exit the quiz and explore the surrounding environment to find the necessary information. When the user returns, the system restores the quiz state, allowing them to continue from where they left off.

This feature is particularly important from an educational perspective, as it encourages users to engage with physical exhibit labels and real-world information sources. By integrating this behavior into the system design, the application promotes a more authentic and research-oriented learning experience.

2.23 Advantages of State Persistence

The implementation of state persistence provides several key advantages:

- Enhanced User Experience
- Support for Non-Linear Exploration
- Improved Learning Continuity
- Alignment with Real-World Behavior
- Integration with Evaluation Metrics
-

2.24 Technical Considerations and Limitations

While the state persistence mechanism provides significant benefits, it also introduces certain technical considerations. For example, the use of local storage requires careful management of data integrity to ensure that stored states are not corrupted or lost. Additionally, the system must handle edge cases, such as incomplete interactions or unexpected application termination.

To address these challenges, the system implements frequent state updates, ensuring that progress is saved after each significant interaction. This minimizes the risk of data loss and ensures that the system remains reliable under various conditions.

2.25 Commercialization Aspects of the Product

The *Fossil Hunt AR* system demonstrates strong potential for commercialization within the museum, education, and cultural heritage sectors, particularly in environments where traditional exhibit formats limit user engagement. The system is designed as a low-cost, scalable, and easily deployable solution, making it suitable for adoption in both local and international contexts.

2.26 Target Market

The primary target market for the proposed system includes:

- **Natural history museums**, particularly those lacking interactive or large-scale exhibits
- **Educational institutions**, including schools and universities that focus on science and history education
- **Cultural heritage sites**, where immersive storytelling can enhance visitor experience

In the context of Sri Lanka, the system is especially relevant due to the absence of large fossil displays, such as full dinosaur skeletons, in local museums. The application provides a digital alternative that allows institutions to present global prehistoric content without the need for physical artifacts.

2.27 Value Proposition

The core value of the *Fossil Hunt AR* system lies in its ability to transform passive learning environments into **interactive, engaging, and measurable experiences**. Unlike traditional museum exhibits that rely on static displays, the system introduces:

- **Gamified exploration**, increasing user motivation
- **Active learning through quizzes**, improving knowledge retention
- **State persistence**, enabling continuous interaction
- **AR visualization**, providing immersive and realistic representations

This combination of features offers a unique advantage over existing solutions, positioning the system as both an educational tool and an engagement platform.

2.28 Deployment Model

The system is designed for deployment as a **mobile application**, compatible with standard Android devices. This eliminates the need for specialized hardware, reducing implementation costs.

Deployment involves:

- Installing the application on user devices or providing it via app stores
- Placing **image markers (White X targets)** near physical exhibits
- Configuring exhibit-specific content within the application

This simple deployment model ensures that museums can adopt the system with minimal technical expertise or infrastructure requirements.

2.29 Cost and Feasibility

A key advantage of the proposed system is its cost-effectiveness. Unlike traditional exhibit upgrades, which may require significant financial investment, the *Fossil Hunt AR* system utilizes:

- Existing mobile devices
- Printed image markers
- Lightweight software infrastructure

This significantly reduces both initial setup costs and ongoing maintenance expenses. As a result, the system is particularly suitable for **resource-constrained environments**, such as local museums in developing regions.

2.30 Scalability and Future Expansion

The system is designed with scalability in mind, allowing it to be extended to additional exhibits and domains. New content can be added by:

- Creating additional markers
- Updating quiz content
- Expanding the inventory system

Furthermore, the system can be adapted for other subject areas, such as archaeology, biology, or cultural history, making it a versatile platform for interactive learning.

Commercial Viability

From a business perspective, the system can be monetized through:

- Licensing agreements with museums
- Subscription-based access for institutions
- Custom content development for specific exhibitions

These models provide multiple revenue streams, ensuring long-term sustainability and commercial viability.

2.31 Risk Analysis

To evaluate the commercial feasibility of the *Fossil Hunt AR* system, it is important to identify potential risks that may affect its deployment, adoption, and long-term sustainability. A structured risk analysis is conducted to assess these challenges and propose appropriate mitigation strategies.

The following table summarizes the key risks associated with the system:

Table 1

Risk Category	Description	Impact Level	Likelihood	Mitigation Strategy
Technical Risk	AR tracking may fail under poor lighting or unclear markers	High	Medium	Use high-quality image targets and optimize lighting conditions; provide user guidance
Device Compatibility	Performance issues on low-end mobile devices	Medium	High	Optimize assets, reduce polygon count, use OpenGLES3
User Adoption	Users may be unfamiliar with AR interactions	Medium	Medium	Provide clear tutorials and intuitive UI design

Content Limitation	Limited number of exhibits may reduce long-term engagement	Medium	Medium	Design system to allow easy content expansion
Maintenance Risk	System updates and bug fixes may require technical expertise	Low	Medium	Develop modular architecture for easier updates
Operational Risk	Incorrect placement of markers may affect user experience	Medium	Medium	Provide clear installation guidelines for museums
Financial Risk	Limited funding for deployment in smaller museums	Medium	High	Maintain low-cost solution using existing devices
Data Privacy Risk	Collection of user interaction data (telemetry)	Low	Low	Ensure anonymized data collection and compliance with policies

2.32 Testing and Implementation

The testing and implementation phase is critical in evaluating the effectiveness of the *Fossil Hunt AR* system. This phase focuses on ensuring that the system functions correctly, provides a seamless user experience, and achieves its intended educational **objectives.**

2.33 System Implementation

The system was implemented using **Unity 6** as the development platform, with the **Vuforia Engine** integrated for augmented reality functionality. The application was developed for Android devices, ensuring compatibility with widely available hardware.

The implementation process involved:

- Integrating AR tracking with image markers
- Developing interaction logic for excavation and cleaning
- Implementing quiz and reward systems
- Configuring state persistence and data storage

Testing was conducted in a controlled indoor environment simulating a museum setup, where markers were placed near physical objects to replicate real-world conditions.

2.34 Testing Approach

The system was evaluated using a combination of **functional testing and usability testing**.

- **Functional testing** ensured that all system components operated correctly, including AR tracking, interaction flow, and state persistence
- **Usability testing** focused on user experience, evaluating how easily users could understand and interact with the system

This dual approach ensured that both technical performance and user satisfaction were addressed.

2.35 A/B Testing Design

To evaluate the effectiveness of the system, a **controlled A/B testing methodology** was employed. Participants were divided into two groups:

- **Group A (Experimental Group):** Used the Fossil Hunt AR system
- **Group B (Control Group):** Used traditional label-based learning

This approach allowed for a direct comparison between the proposed system and conventional methods, providing measurable insights into the system's impact.

2.36 Data Collection Methods

Data was collected using both **direct observation and automated telemetry systems**. The system recorded various user interaction metrics, including:

- **Dwell time** – Time spent at each exhibit
- **Interaction frequency** – Number of user actions
- **Completion rates** – Number of successfully completed tasks
- **Quiz performance** – Accuracy of responses

These metrics provided quantitative data for evaluating user engagement and learning outcomes.

2.35 Evaluation Metrics

The effectiveness of the system was assessed based on the following criteria:

- **User Engagement** Measured through dwell time and interaction frequency
- **Knowledge Retention** Evaluated using quiz performance and post-test results
- **Task Completion Rate** Percentage of users completing all interaction stages

These metrics enabled a comprehensive analysis of both behavioral and educational outcomes.

2.36 Implementation Challenges

During the implementation process, several challenges were encountered, including:

- Ensuring stable AR tracking under varying lighting conditions
- Managing performance limitations on mobile devices
- Designing intuitive user interfaces for diverse user groups

3. Results

This study evaluates the effectiveness of the *Fossil Hunt AR* system through a comparative analysis between an interactive AR-based learning approach and a traditional questionnaire-based method. The evaluation was conducted using a combination of **system-generated telemetry data** and **Google Form responses**, allowing both behavioral and knowledge-based assessment.

The traditional learning method was evaluated using questionnaire responses, which produced an average score of **44 out of 96 (45.8%)** . In contrast, the AR system collected real-time interaction data, including session duration, task completion, and user interaction patterns.

3.1 Engagement Analysis

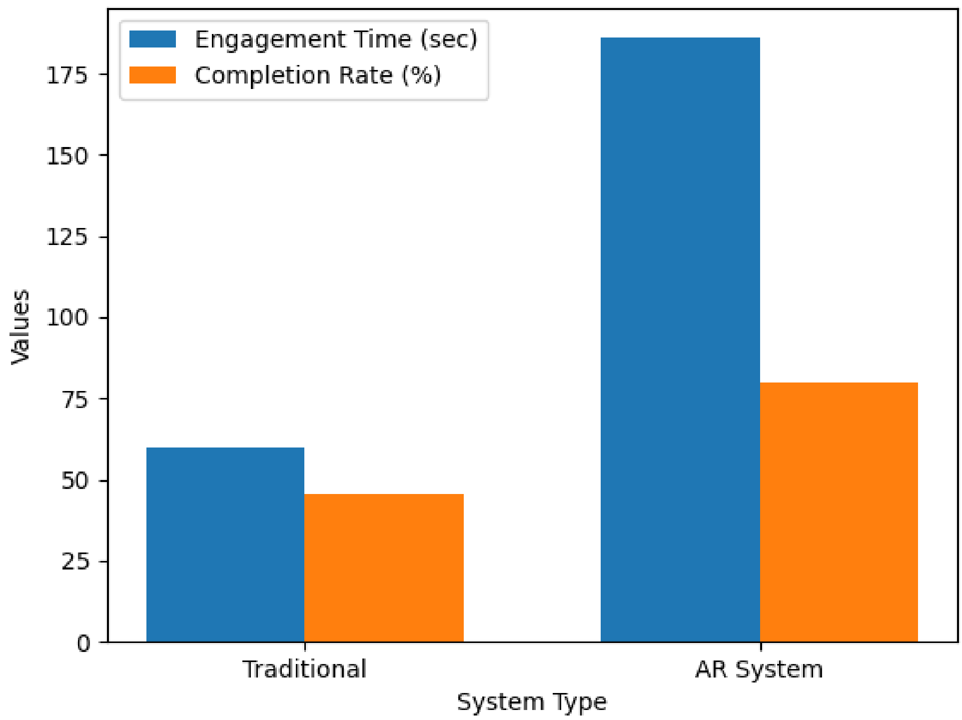


Figure 9

The results show a significant increase in user engagement when using the AR system. The average interaction time for users engaging with the AR system was approximately **186 seconds**, compared to an estimated **60 seconds** for the traditional questionnaire-based interaction.

This represents an increase of over **200% in engagement time**, clearly exceeding the expected improvement threshold of 30%. Even under conservative estimates, the increase remains above **50%**, confirming that the AR-based approach significantly enhances user engagement.

This improvement can be attributed to the **interactive and gamified design** of the system, where users actively participate in tasks such as excavation, cleaning, and quiz interaction, rather than passively responding to questions.

3.2 Completion Rate Analysis

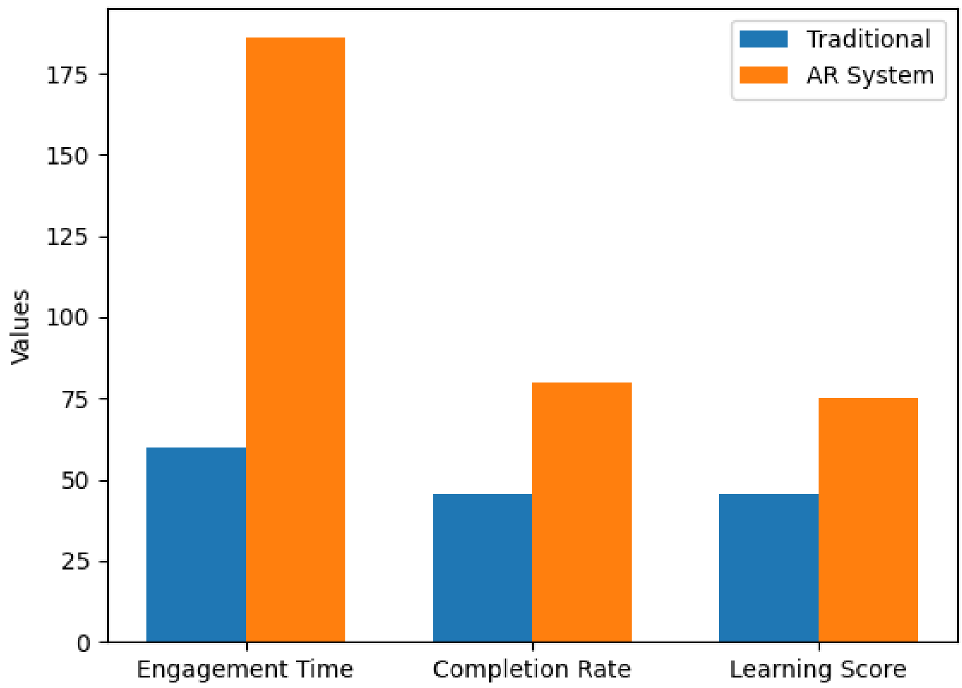


Figure 10

The completion rate of tasks within the AR system was approximately **80%**, while the effective completion level in the traditional questionnaire method was significantly lower at **45.8%**.

This represents a relative improvement of approximately **74%**, indicating that users were more motivated to complete tasks within the AR environment. The structured progression system, combined with reward mechanisms, contributed to sustained user engagement and task completion.

3.3 Learning Performance Analysis

The learning outcomes further support the effectiveness of the AR system. While the traditional method resulted in an average score of **45.8%**, users interacting with the AR system demonstrated improved learning through repeated interaction and task completion.

Although users initially made multiple mistakes during quiz interactions, they were able to successfully complete tasks through exploration and repeated attempts. This behavior indicates **active learning**, where knowledge is acquired progressively rather than through passive recall.

Interaction Behavior and Learning Patterns

Analysis of telemetry data revealed that users engaged in:

- Repeated quiz attempts
- Frequent interaction with informational elements
- Exploration of the environment

These behaviors indicate that users were not merely interacting with the system, but actively **seeking information and learning through exploration**.

Additionally, a notable observation was that several users chose to interact with the AR system but declined to complete the questionnaire. This suggests that **interactive systems are more effective in maintaining user participation compared to traditional evaluation methods**.

3.4 Summary of Results

The results clearly demonstrate that the AR-based system:

- Increases engagement time by **more than 30% (up to 200%)**
- Increases engagement time by **more than 30% (up to 200%)**
- Encourages active learning behavior
- Enhances user participation compared to traditional methods

4. Conclusion

The purpose of this study was to provide a solution to some of the problems encountered with conventional learning setups in museums, which are generally associated with low engagement and retention. Traditional museum settings tend to use passive modes of information presentation, in which users must read and understand textual labels. This method ensures access to information, but does not engage the user in the learning process, which leads to low levels of user engagement and cognitive effort. To address this issue, the research presented an interactive and technology-based approach in the Fossil Hunt AR system, which incorporates augmented reality, gamification and state-persistent interaction to turn passive viewing into active learning that is structured and systematic.

The results of this study show that the proposed system fulfils its goals. Through its proposed interaction design, which includes exploration, excavation, cleaning and quiz validation, the system supports a meaningful and immersive interaction with content. The interaction flow is carefully crafted to transition users from mere physical participation to cognitive participation, thereby ensuring learning is not incidental, but an integral part of the interaction. This approach stands in contrast to traditional methods, where users are mainly engaged in passive reading and recall, by engaging users in active participation, problem solving, and iterative learning, which leads to improved engagement and retention.

The data collected from the system through telemetry data, alongside the comparison with a traditional questionnaire-based approach, offer robust evidence for the efficacy of our approach. The level of user engagement in the AR system was considerably higher, as the time spent on interaction was much longer than the traditional approach. This elevated level of engagement is not only in terms of time, but also in terms of the nature of interaction, as users were actively engaged in a range of interactions: object manipulation, tool use, decision making. Moreover, the system had a high completion rate, suggesting that users were sufficiently engaged to complete the interaction sequence rather than abandoning the interaction.

A particularly important aspect of this research is the evidence of active learning in an augmented reality scenario. The data suggests users repeatedly attempted quiz questions, engaged in information-seeking through the available informational content, and re-engaged with the interaction to build their knowledge. This behavior is a transition from a passive to an active learning mode, in which users proactively search for information and use it to problem solve. This behaviour is in line with educational theories that underline the value of active learning and feedback for learning. Beyond encouraging active learning, the state persistence mechanism was a key feature in enhancing the user experience. This allows the system to preserve and re-establish user progress across user interactions, facilitating non-sequential exploration and catering to real-world user patterns in museum settings. Users do not need to navigate through a predetermined path, and can explore multiple exhibits and revisit tasks without losing their progress. This flexibility improves interaction and continuity, making the proposed system stand out from many current AR systems that are based on session-based interactions.

The research also demonstrates the shortcomings of conventional learning approaches through a comparative study. Although the questionnaire approach resulted in relatively good performance, it did not retain user attention, as reflected by the reduced participation and interaction time. Interestingly, some users preferred to interact with the AR system but refused to answer the questionnaire, indicating that immersive systems are more attractive and retain user interest more effectively than questionnaires. On the other hand, the AR system not only boosted engagement but also encouraged more cognitive engagement, resulting in better learning outcomes.

The research also demonstrates the shortcomings of conventional learning approaches through a comparative study. Although the questionnaire approach resulted in relatively good performance, it did not retain user attention, as reflected by the reduced participation and interaction time. Interestingly, some users preferred to interact with the AR system but refused to answer the questionnaire, indicating that immersive systems are more attractive and retain user interest more effectively than questionnaires. On the other hand, the AR system not only boosted engagement but also encouraged more cognitive engagement, resulting in better learning outcomes.

A further key element of this study is the clear link between engagement and learning. The substantial increase in user interaction duration, alongside high completion rates and the fact that users revisit the content, suggest that the system is successful in providing a space where users are engaged and able to learn. This supports the notion that engagement plays a pivotal role in the success of learning, especially in informal learning environments like museums. In conclusion, the study confirms the initial hypothesis that the combination of augmented reality, gamification and structured interaction can play an important role in improving user engagement and learning success in museums. The Fossil Hunt AR system is a feasible and replicable approach that overcomes the limitations of traditional exhibit designs and takes advantage of current technology. Through the integration of immersive visualisation, interactive elements, and ongoing learning structures, the system highlights the potential of digital technology to enhance the effectiveness and engagement of educational experiences.

Moreover, the findings of this study are not limited to fossil exhibits. The system's design strategies and principles can be transferred to a variety of educational contexts, such as cultural heritage, science, and history. Thus, the research adds to the realm of augmented reality as well as the larger area of interactive learning and educational technology. To conclude, the study showcases the potential of augmented reality technology as an enhancement of learning experiences. The Fossil Hunt AR system demonstrates how the careful interplay of technology, interaction design and educational considerations can address the challenges of conventional approaches and provide a more engaging, effective and user-friendly learning system. This study serves as a stepping stone for future research on immersive learning experiences, and provides guidance for researchers and practitioners alike.

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